Chris Calvin Demta

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Full-stack AI engineer who ships AI products end to end: a LangGraph HR agent serving 1,500+ employees (70% fewer SAP tickets) and a questionnaire assistant that cut a 3 to 4 day client process to about 2 hours. Now building product-grade agent systems outside consulting. Published Machine Learning researcher with a research internship at Illinois Institute of Technology.

EXPERIENCE

AI Engineer (Associate), PricewaterhouseCoopers Sep 2024 – Present
Hyderabad

Project: Enterprise HR & SAP Support Agent (LangGraph, GPT-4o, Azure AI Search, SuccessFactors API, MS Teams)

- Cut SAP ticket volume 70% with a multi-agent HR assistant (LangGraph) deployed to 1,500+ employees across MS Teams and web chat; handles 150+ daily queries with no human intervention.
- Built Azure AI Search retrieval with HNSW indexing and cross-encoder reranking over 6,000+ pages of HR policy docs, personalized per employee via live SuccessFactors calls (salary band, leave balance, role).
- Improved answer faithfulness ~30% over a naive-chunking baseline by engineering the retrieval pipeline on OpenAI text-embedding-3-small with recursive text splitting.

Project: AI Questionnaire Prefill System — full-stack (Python, FastAPI, LangChain, Claude on Bedrock, Angular, TypeScript, SQL Server, Salesforce)

- Cut the time to complete a client engagement risk questionnaire from 3–4 days to about 2 hours by building an AI assistant that prefills ~600 questions from the client's contract and engagement data; users now only review and verify before submitting.
- Made every AI answer verifiable: the interface shows where each answer's data came from, why it was chosen, and a confidence score, and users changed only ~5–6 answers per questionnaire in the audit trail.
- Designed the AI workflow end to end: pulls data from five enterprise systems in parallel, answers in batches, and re-checks lower-confidence answers; rule-based shortcuts handle straightforward fields to reduce AI usage.
- Built a rules engine that shows or hides questions as answers change, so reviewers only see and answer what is relevant to their engagement.
- Delivered production-ready: human review before submission, save-and-resume, a full audit trail of every action for traceability, and direct submission into Salesforce.

Project: Pharmaceutical NLU Chatbot & Document Intelligence (Azure Language Studio, pgvector, ABBYY OCR, LangChain)

- Fine-tuned Azure NLU models across 40+ intents / 200+ entities, improving intent classification precision ~18% over baseline; owned model registry, versioning, and CI/CD across dev/staging/prod.
- Built a production RAG pipeline (PostgreSQL/pgvector) over 8,000+ pages of pharma documents, cutting manual lookup time ~65%.
- Applied ABBYY OCR for structured invoice field extraction, automating a manual workflow at >90% field accuracy.

Technology Consultant Intern, PricewaterhouseCoopers Jan 2024 – Sep 2024
Hyderabad

- Built a client-facing chatbot on OpenAI's API, cutting average Tier-1 support response time ~40%.
- Built Power BI dashboards surfacing client data trends that directly informed quarterly strategy for a pharma-sector client.

AI Research Intern, Illinois Institute of Technology — Prof. Jafar Saniie May 2023 – Jul 2023
Chicago

- Selected from 800+ undergraduates for ISRE '23; awarded $7,300+ in combined BIT Mesra / IIT Chicago scholarships.
- Built and benchmarked three music-generation architectures (LSTM, LSTM+Transformer-XL, LSTM+Transformer-XL+FastAI) on MIDI piano data, validated via human evaluation.
- Built image/video processing pipelines on Jetson Nano for low-power IoT deployment.

PORTFOLIO PROJECTS

FinSight — Autonomous Financial Research Agent June – July 2026
GitHub: https://github.com/calv8n/FinsightAgent

- Built a CodeAct-style agent from scratch (no LangChain/LangGraph) that answers financial questions by autonomously retrieving SEC 10-K filings, writing Python, and executing it in a hardened Docker sandbox.
- Built a hybrid RAG pipeline: SEC EDGAR ingestion → item-header chunking → MiniLM embeddings → Qdrant → BM25 sparse search → RRF fusion → cross-encoder reranking.
- Built a 4-layer sandboxed code executor (network isolation, capability drop, builtins whitelist, syscall auditing) — all four layers must fail simultaneously for unsafe code to execute.
- Engineered a 5-model fallback chain across Groq models with retry/backoff, handling rate limits without dropping requests.

- Built a FastAPI backend with SSE streaming and persistent session state, and a Streamlit frontend with real-time thought/code/answer display.
- Evaluated on a 20-question financial Q&A benchmark (DCF, LBO, Monte Carlo, regression) scored by LLM-as-judge.

EDUCATION

B.Tech, Computer Science and Engineering — Birla Institute of Technology, Mesra May 2024

GPA: 7.82 / 10

SKILLS

Languages: Python, SQL, TypeScript, JavaScript

ML/AI: LLMs, Transformers, Fine-tuning (LoRA/QLoRA), RAG, Agentic Design Patterns, Multi-Step Agents, Function/Tool Calling, Prompt Engineering, Human-in-the-Loop Workflows, Intent Recognition, Entity Extraction, NLP, GANs, RNNs/LSTMs, LLM Evaluation & Observability, CoT

Frameworks: LangChain, LangGraph, LlamaIndex, HuggingFace, PyTorch, TensorFlow, Scikit-learn, FastAPI, SQLAlchemy, DSPy

Frontend: Angular, PrimeNG, HTML/CSS, REST API integration

Cloud & MLOps: Azure (AI Search, Cosmos DB, Functions, AI Foundry, Language Studio), Docker, Git, CI/CD, pgvector/PostgreSQL

Retrieval: HNSW indexing, Cross-encoder reranking, Hybrid (BM25 + dense) search, RRF fusion

PUBLICATIONS

Mondal, D., Demta, C.C., Bhattacharjee, V. “Application of GANs for Generation of Handwritten Digits and Letters.” ICMIA 2023, BIT Lalpur.

CERTIFICATIONS

Anthropic Developer Foundations • Microsoft Power Automate (PL-500) • Kore.ai Advanced Certification